Remote Terminal Unit Database

Participants are instructed to create following tables with appropriate data types, primary key, foreign key, not null and check constraints.

Insert data as shown below into respective tables

Table - RTU

RTU_ID	LOCATION	IP_ADDRESS
R101	BENGALURU	192.168.0.1
R102	MUMBAI	192.168.0.2
R103	CHENNAI	192.168.0.3
R104	NEW DELHI	192.168.0.4
R105	KOLKATA	192.168.0.5

Table – SENSORS

SENSOR_ID	RTU_ID	STATUS	SENSOR_TYPE	REQ_VAL	CURR_VAL
S801	R101	ENABLED	PRESSURE	34	33
S802	R101	ENABLED	TEMPERATURE	98	90
S803	R101	DISABLED	TEMPERATURE	96	
S804	R102	ENABLED	PRESSURE	30	32
S805	R102	ENABLED	PRESSURE	36	32
S806	R102	DISABLED	TEMPERATURE	108	
S807	R102	ENABLED	BAROMETER	750	660
S808	R103	ENABLED	BAROMETER	780	690
S809	R103	ENABLED	ULTRASONIC	510	450
S810	R104	DISABLED	ULTRASONIC	530	
S811	R104	ENABLED	PRESSURE	60	50

Table – USERS

USER_ID	USER_NAME	ROLE
U501	PRAKASH	MONITOR
U502	PARTHIV	ADMINISTER
U503	SMITHA	ADMINISTER

Table – CONTROLLER

USER_ID	SENSOR_ID
U501	S801
U501	S804
U501	S806
U502	S802
U502	S804
U503	S805
U503	S806
U501	S809
U502	S810
U503	S811
U501	S807
U503	S808
U502	S801
U501	S808

Write following queries based on above tables

Simple Queries

1. Fetch details of all the sensors which are disabled
2. List sensors which are having required value less than current value
3. List sensors which are of type TEMPERATURE and required value greater than 96. Sort on required value field.
4. List sensors which has no current values
5. List all sensors which are of type TEMPERATURE & PRESSURE.
6. List sensors with rtu_id 102 and sensor type pressure and required value greater than 30.

Single Row Functions

1. Write a query which will display username in upper case, role first letter capitalized and all other letters lower case and length of the username.
2. Display percentage of difference between req_value and curr_value as a whole number.

Group Functions

1. List how many sensors are present of type PRESSURE.
2. Display number of sensors enabled and disabled. Sort on number of sensors
3. List how many sensors are present in each sensor type.
4. List number of sensors under each RTU_ID.
5. Display how many sensors are controlled by each user. Exclude users who are controlling less than 3 sensors and also restrict users with id “U503”. Sort on user_id.

JOINS

1. Display RTU details alongwith sensor details for all RTU's
2. Display user details and sensor details controlled by each user.
3. Display user details and sensor details controlled by each user alongwith RTU details.
4. List how many sensors are controlled by each user. Display user_name and count. Sort on count.
5. Display user details who are controlling sensor_type TEMPERATURE
6. Display user details that is controlling sensor S801 along with sensor details and location.

SUB-QUERIES

1. Display sensor details located in CHENNAI
2. Display user details controlling sensors in MUMBAI location
3. Display sensor details controlled by SMITHA
4. Display user details who are controlling sensors with id between S803 and S808
5. Display user details that are controlling most number of sensors.
6. Display RTU details which has most number of sensors.